Coursework Brief

Task To Be Done

You are given the ML 2026A2 Hurricane dataset (access via this link) to build one multilayer perceptron model that forecasts four hurricane-relevant measures (rain precipitation rate, wind speed, wind direction, surface air pressure) a week into the future.

For example, assuming the current date were 18 Dec 1965, your model would be able to forecast rain precipitation rate, wind speed, wind direction, and air pressure for 25 Dec 1965 for any location, e.g. latitude 50.87, longitude -0.087 (at University of Sussex).

For more information about the dataset, see the README file included in the dataset folder.

Your multilayer perceptron model must be built using the PyTorch library.

What To Submit

You are to submit 3 separate documents as specified below. Do NOT submit them as a zipped folder.

1. (90 marks) **Report** (submit as a pdf file) **–** The report must have EXACTLY 4 sections in the order given below. Also see below for what you are expected to report in each section.

   A. (max - 35 marks) **Features & Labels**
      - List the specific input data (features) and output data (labels) of your model including their dimensionalities, and provide valid justification for your features and labels selection. Also, describe very clearly how each feature and label was extracted from the given data, with clear rationale for the approach taken.

   B. (max - 10 marks) **Preprocessing**
      - List and describe very clearly any (other) preprocessing that you did on your data toward building your model, with clear rationale provided in each case.

   C. (max - 25 marks) **Model**
      - Describe your model very clearly, including its hyperparameters, how it was optimized, and a simple diagram that illustrates your model architecture. Make sure the diagram specifies your own model architecture and not a generic multilayer perceptron. Clarify the rationale for your multilayer perceptron.
      - List and describe very clearly the steps you took to prevent overfitting for your model. Note that 2 marks will be deducted for incorrectly including strategies that do not prevent overfitting.

   D. (max - 20 marks) **Generalizability**
      - Report the generalizability of your model, with clear and valid rationale provided for any performance metric reported. Also, describe how each performance value was computed from your model output.

- State the total number of data instances you used for building and evaluating your model, and describe very clearly and with valid rationale how the data was split to assess the generalizability of your model.

2. (5 marks) **Code notebook** (submit as a ipynb file) - Prepare the code used to complete all the tasks above and submit as a single ipynb file that can be run. ONLY include code for details specified in your report. Code for any extra exploration etc should be removed; otherwise, your code would seem to not match your report.

3. (5 marks) **Model outputs for the test set(s)** (submit as a csv file) - Prepare the test set outputs of your model in a single csv file. The file should have an appropriate header row and two columns per label type (1 for the true value & 1 for the corresponding model prediction).

Other Important Notice

1. Your report must NOT be longer than 10 pages. Also, it must NOT have more than 3 tables and 3 figures. Nothing after p. 10 in the report will be read or marked. Also, tables after the first 3 and figures after the first 3 will be ignored.

2. You are not required to code a multilayer perceptron from its math principle and are allowed to use standard machine learning libraries. However, you must ONLY use the PyTorch library for implementing your multilayer perceptron model.

3. You are NOT permitted to use or submit someone else's code, output, or report as yours. You are only allowed to use code snippets from the lab materials given by the tutor, from software library documentations as stated in #2 above, or from recommended textbooks.

4. You must NOT use generative artificial intelligence (AI) to generate any materials or content for your assessment submissions.

   Note that:

   - The baseline position at the University is Sussex is that the use of generative AI material in assessment submissions is prohibited, unless explicitly permitted by the module convenor.
   - Students registered with the Disability Advice team and in receipt of reasonable adjustments are still permitted to use other assistive technology as required.
   - If in any doubt about what is permissible, students should check with the module convenor.

Link to learning outcomes

This assessment is designed to evaluate how well you have achieved the module's learning outcomes. See Module information: Machine Learning [25/26] for the learning outcomes.